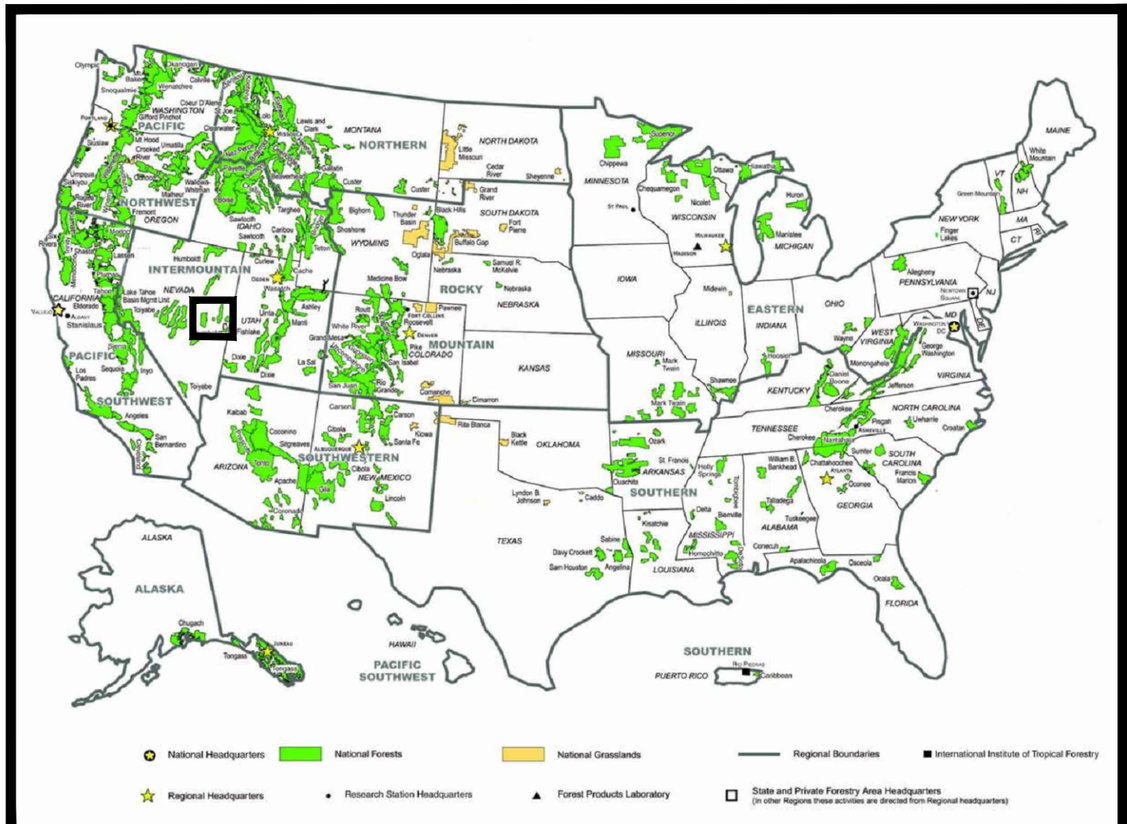
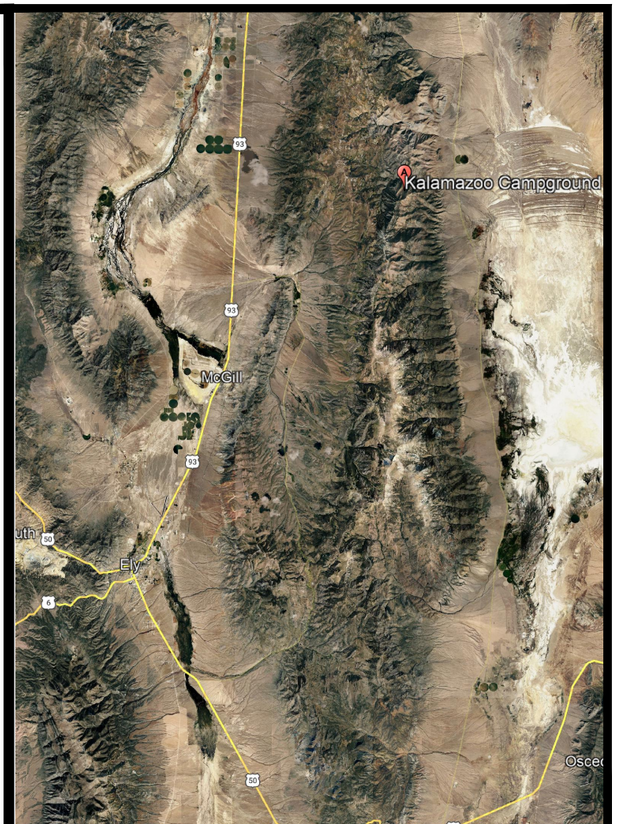
United States Department of Agriculture
Forest Service

WHITE PINE COUNTY, NEVADA
INTERMOUNTAIN REGION
HUMBOLDT-TOIYABE NATIONAL FOREST
ELY RANGER DISTRICT

KALAMAZOO FENCE



FOREST LOCATION MAP



PROJECT LOCATION ~~MAP~~

TRAVEL DIRECTIONS: FROM ELYRANGER DISTRICT OFFICE TRAVEL SOUTH OF US-50 (GREAT BASIN BLVD) FOR APPROX. 28 MILES. TURN LEFT NV-893 NORTH FOR APPROX. 38 MILES. TURN LEFT ONTO KALAMAZOO ROAD FOR APPROX. 3.5 MILES TO KALAMAZOO CAMPGROUND.

INDEX OF SHEETS

SHEET	SHEET TITLE
G-1	COVERSHEET
D-1	DETAILS: FENCE LINE LOCATION
D-2	GENERAL NOTES AND SPECIFICATIONS
D-3	DETAILS: FENCE LINE CONSTRUCTION
D-4	DETAILS: GATE CONSTRUCTION

ALL WORK SHALL CONFORM TO FP-24 STANDARD SPECIFICATIONS FOR CONSTRUCTION OF ROADS AND BRIDGES ON FEDERAL HIGHWAY PROJECTS, US DEPARTMENT OF TRANSPORTATION, FEDERAL HIGHWAY ADMINISTRATION

APPROVED:

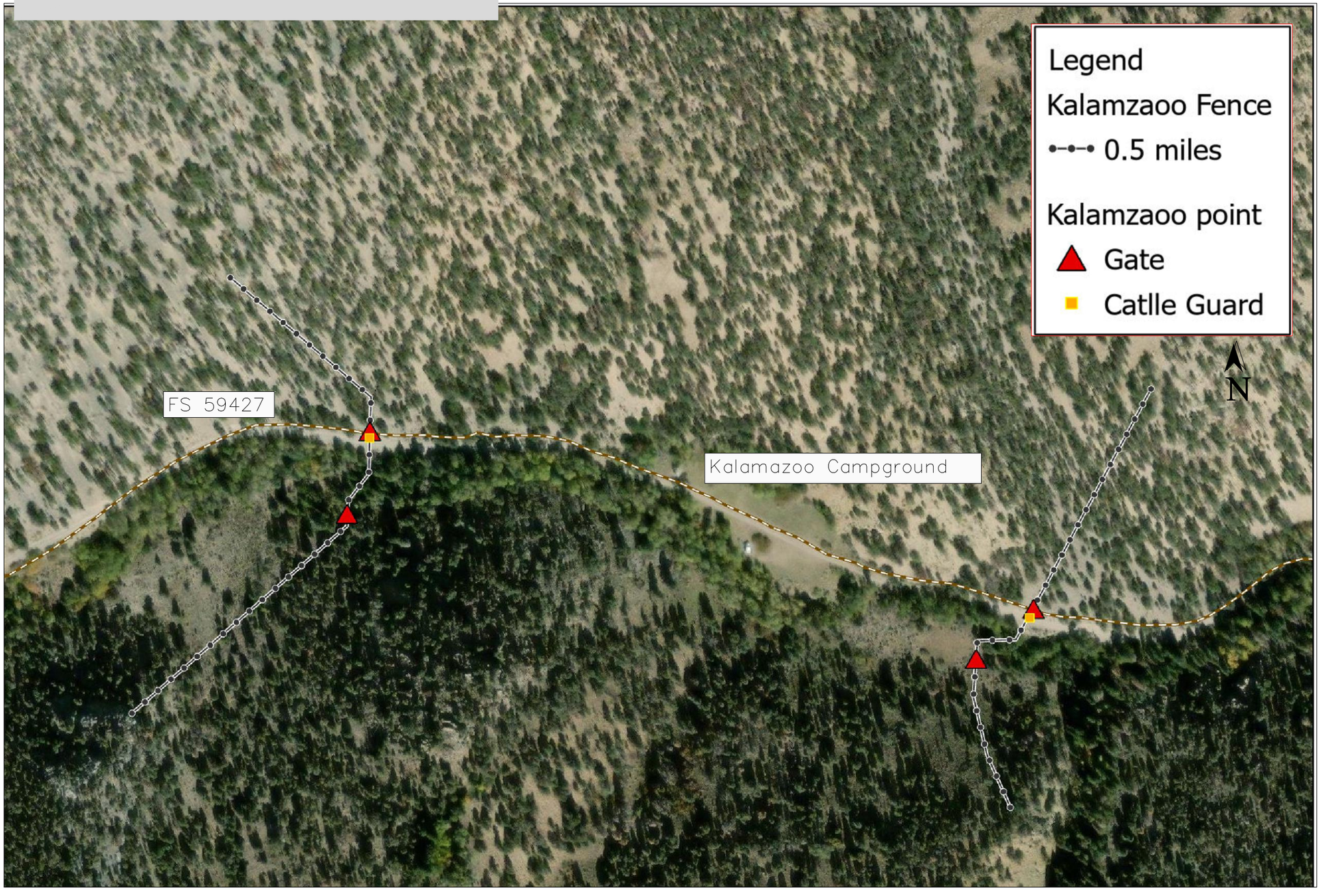
KEVIN WILMOT Digitally signed by KEVIN WILMOT
Date: 2026.07.08 12:43:58 -07'00'

FOREST ENGINEER

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NOTES:
- THERE IS NO EXISTING FENCE LINE TO FOLLOW
- INSTALL WIRE GATES
  * (ESTIMATED 4 TOTAL GATES)
- RUN AND ANCHOR EACH STRETCH OF NEW FENCE LINE



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PROJECT NAME

KALAMAZOO FENCE

HUMBOLDT-TOIYABE NATIONAL FOREST

ELY DISTRICT

DRAWING TITLE

DETAILS: KALAMAZOO FENCE LINE LOCATION

DATE 2/19/2026	ARCHIVE NO.
DESIGNER C.CUBIC	DRAWING SHEET NO. D-1
DRAWN C.CUBIC	
CHECKED K. WILMOT	
PROJECT NO.	SHEET 2 OF 5

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GENERAL NOTES

MEASUREMENT AND PAYMENT
- BOTH THE REMOVAL OF ROUGHLY 13.1 MILES OF BURNED AND UNBURNED EXISTING FENCE AND MOBILIZATION WILL BE PAID FOR AS LUMP SUMS.
- PAYMENT FOR WIRE FENCE WILL BE MEASURED BY LINEAR MILE BETWEEN CENTERS OF END POSTS, INCLUDING THE LENGTH BETWEEN CENTERS OF GATE POSTS, ALONG THE SLOPE OF THE BOTTOM WIRE COMPLETED IN PLACE AND ACCEPTED.

MATERIAL REMOVAL
- WIRE, T–POSTS, AND OTHER OLD FENCING MATERIAL MUST BE REMOVED FROM THE PROJECT SITE AND HAULED TO AN APPROPRIATE LEGAL DISPOSAL LOCATION OFF OF FS LAND.
- CLEARED TREES AND DEBRIS MAY BE SCATTERED AWAY FROM THE FENCELINE ON SITE ONCE ALL WIRE HAS BEEN REMOVED.
- VEGETATION ALONG THE ROUTE OF THE FENCE LINE MAY BE CLEARED ONLY AS NEEDED TO ENSURE THAT THE FENCE IS STRAIGHT BETWEEN BRACES.
- MATERIAL IS TO BE REMOVED AT GROUND LEVEL WITH MINIMUM DISTURBANCE TO SOIL SURFACE TO PROTECT AESTHETICS AND PREVENT EROSION.

MATERIAL SPECIFICATIONS

LINE POSTS
- STANDARD T–POSTS SHALL CONFORM TO AASHTO M281, WEIGHING NOT LESS THAN 1.33 POUNDS PER FOOT OF LENGTH EXCLUSIVE OF ANCHOR PLATE.
- T–POSTS SHALL BE AT LEAST 6–FOOT IN LENGTH.
- STEEL POSTS ARE TO BE STUDDED, EMBOSSED OR PUNCHED FOR THE ATTACHMENT OF WIRE AND HAVE AN ANCHOR PLATE NEAR THE BOTTOM OF THE POST.
- T–POSTS SHALL BE FACTORY PAINTED GREEN WITH A GRAY OR LIME GREEN TOP.

WIRE
- BARBED WIRE SHALL BE TWO STRANDS OF NO. 12–GAUGE ZINC–COATED STEEL WIRE WRAPPED AROUND EACH OTHER WITH 4–POINT, NO. 14–GAUGE W BARBS SPACED NOT MORE THAN FIVE INCHES APART CONFORMING TO AASHTO M280.
- SMOOTH WIRE FOR BOTTOM FENCE STRAND SHALL BE TWO STRANDS OF NO. 12–GAUGE ZINC–COATED STEEL WIRE WRAPPED AROUND EACH OTHER

STAYS
- STAYS ALONG THE FENCE LINE SHALL BE NO. 9 WIRE MANUFACTURED WIRE STAYS
- STAYS ALONG THE GATE SHALL BE EITHER PEELED JUNIPER, CEDAR, OR CYPRESS AND HAVE A 2– TO 3–INCH DIAMETER.
- WOODEN STAYS SHOULD BE LONG ENOUGH TO EXTEND AT LEAST 4–INCHES ABOVE THE TOP WIRE STRAND.

CLIPS
- FENCE LINE WIRE SHALL BE FASTENED TO T–POSTS USING NO. 12–1/2–GAUGE GALVANIZED STEEL WIRE FASTENERS MANUFACTURED FOR THIS PURPOSE.

OBJECT MARKERS
- SEE GATE DETAIL (D–3)

BRACE POSTS – STEEL
- STEEL POSTS AND BRACES SHALL BE GALVANIZED SCHEDULE 40 PIPE. USED DRILL STEM PIPE MEETING DIAMETER REQUIREMENTS MAY BE USED IN PLACE OF GALVANIZED STEEL POSTS CONTINGENT UPON FS APPROVAL
- ALL METAL CORNER, GATE AND IN–LINE VERTICAL BRACE UNIT POSTS SHALL BE PIPE AT LEAST 7–FEET IN LENGTH WITH A 2.375–INCH OUTSIDE DIAMETER.
- HORIZONTAL OR DIAGONAL CROSS BRACES WILL BE NEW AND SHALL HAVE A MINIMUM 2.375–INCH OUTSIDE DIAMETER.
- STEEL HORIZONTAL BRACES WILL BE A MINIMUM OF 5–FEET IN LENGTH.
- STEEL DIAGONAL CROSS BRACES WILL BE A MINIMUM OF 5½–FEET IN LENGTH.

BRACING WIRE
- TENSION WIRES SHALL HAVE A TENSILE STRENGTH NOT LESS THAN 58,000 POUNDS PER SQUARE INCH

WIRE SPACING
- THE BOTTOM WIRE WILL BE SMOOTH, THE TOP THREE WIRES BARBED, WITH SPACING AS SHOWN IN THE TABLE BELOW:

WIRE SPACING FROM GROUND			
BOTTOM WIRE (SMOOTH)	2ND WIRE (BARBED)	3RD WIRE (BARBED)	TOP WIRE (BARBED)
18"	24"	30"	42"

CONSTRUCTION SPECIFICATIONS

GENERAL
- GRADING ALONG FENCE LINE IS NOT REQUIRED. FENCE SHALL GENERALLY BE ERECTED ON UNDISTURBED, NATURAL GROUND.
- FENCE WIRE SHALL NOT BE ATTACHED TO TREES.
- WIRE FENCES SHALL BE REASONABLY STRAIGHT AND MAY NOT DEVIATE MORE THAN 12–INCHES FROM A STRAIGHT LINE BETWEEN ANY CORNER AND BRACE ASSEMBLIES.
- WHEN A FENCE LINE APPROACHES A DITCH, GULLY, OR DEPRESSION THE LAST POST ON LEVEL GROUND SHALL BE PLACED CLOSE ENOUGH TO THE EDGE THAT THE WIRE MAY BE STRUNG TO A POST IN THE DEPRESSION WITHOUT TOUCHING THE GROUND.
- INSTALL SAG BRACING WHEN THE DEPTH OF A DEPRESSION EXCEEDS THE TOTAL VERTICAL WIRE SPACING OVER A MAXIMUM HORIZONTAL RUN OF TWO SPACES.
- END BRACES SHALL BE CONSTRUCTED AGAINST ROCKS OR OTHER BARRIERS IN THE FENCE LINE IN SUCH A MANNER THAT A COW CAN NOT PASS THROUGH.
- PROPOSED WIRE CRIBS/ ROCK JACKS ARE NEGOTIABLE UPON APPROVAL OF DESIGN BY COR

LINE POSTS
- T–POST INTERVALS SHALL BE 12–FEET MAXIMUM
- STAYS SHALL BE PLACED HALFWAY (TYPICALLY 6 FEET) BETWEEN TWO T–POSTS.
- ALL POSTS SHALL BE SET SOLIDLY IN SOIL TO DEPTHS SHOWN ON DRAWINGS. THE TOP OF T–POST ANCHOR PLATES SHALL BE EMBEDDED 4–INCHES MINIMUM.

FENCE HEIGHT
- THE MINIMUM HEIGHT (MEASURED FROM THE GROUND AT POST LOCATIONS) TO THE TOP WIRE SHALL NOT BE LESS THAN 42 INCHES.

LINE WIRE
- WIRE SHALL NEVER BE KINKED OR NICKED, AND SHALL BE PULLED TIGHT WITH DUE CONSIDERATION FOR CONTRACTION AND EXPANSION.
- THE FENCING WIRE SHALL BE PLACED ON THE SIDE OF THE POST EXPECTED TO RECEIVE THE GREATEST PRESSURE
- THE FENCING WIRE SHALL BE FASTENED TO STANDARD T–POSTS BY MEANS OF STEEL WIRE CLIPS MANUFACTURED FOR THIS PURPOSE
- THE TOP LINE WIRE SHALL BE SET SO THAT FENCE POSTS EXTEND A MINIMUM OF 2–INCHES ABOVE THE WIRE.
- ALL LINE WIRES SHALL BE DEAD–ENDED ON THE ANCHOR POST (PULL POST) OF GATE, CORNER, AND IN–LINE BRACE ASSEMBLIES. WIRE ENDS ARE TRIPLE WRAPPED AROUND ANCHOR POSTS (PULL POSTS) AND TWISTED BACK ON THE STRETCHED LINE WIRE WITH AT LEAST SIX TIGHTLY WOUND WRAPS.

CORNER, BRACE, AND GATE POSTS
- BRACES ARE REQUIRED AT ALL CORNERS, GATES, CHANGES IN ALIGNMENT TO THE LINE FENCE, AND AT ALL DEFINITE SLOPE BREAKS.
- LINE BRACES SHALL BE LOCATED AT BOTH SIDES OF THE GATE.
- BRACING SHALL BE ATTACHED TO POSTS USING CONNECTIONS MANUFACTURED FOR THIS PURPOSE.
- TENSION WIRES SHALL BE CONNECTED TO EACH POST USING APPROPRIATE FITTINGS TO PREVENT VERTICAL MOVEMENT AND WRAPPED ON ITSELF A MINIMUM OF 6 TIMES.. THEY WILL BE TWISTED TIGHT WITH THE TENSIONING STICKS LEFT INSTALLED.
- CAP ALL POSTS OR FILL TO THE TOP WITH DIRT OR SAND.
- IN STRAIGHT SECTIONS ON MODERATE TERRAIN, IN–LINE BRACE UNITS ARE REQUIRED AT INTERVALS NOT TO EXCEED 1,320 FEET (110 RODS).
- CORNER BRACE ASSEMBLIES SHALL BE INSTALLED AT ALL POINTS WHERE THE FENCE ALIGNMENT CHANGES 20–DEGREES OR MORE.
- BRACE UNITS ARE REQUIRED AT ANY POINT WHERE THE VERTICAL ANGLE DESCRIBED BY TWO ADJACENT REACHES OF WIRE IS UPWARD AND EXCEEDS 10–DEGREES.
- ANCHOR POSTS SHALL BE SET WITH A 1– TO 2–INCH LEAN AWAY FROM THE DIRECTION OF FENCE PULL.

LINE WIRE TENSION
- LINE WIRES SHALL BE STRETCHED TIGHT. FENCE WIRES SHOULD BE TENSIONED WORKING FROM THE TOP WIRE DOWN.
- TEMPERATURE VARIATIONS MUST BE CONSIDERED WHEN SETTING THE TENSION ON LINE WIRES (WIRE WILL TIGHTEN IN COLD WEATHER AND EXPAND IN HOT WEATHER).

WIRE SPLICING
- WIRE SHALL BE SPLICED BY MEANS OF A WESTERN UNION SPLICE OR BY SUITABLE SPLICE SLEEVES APPLIED WITH A TOOL DESIGNED FOR THAT PURPOSE. THE WESTERN UNION SPLICE SHALL HAVE NO LESS THAT SIX (6) WRAPS OF EACH END ABOUT THE OTHER. ALL WRAPS SHALL BE TIGHTLY WOUND AND CLOSELY SPACED.
- SPLICES MADE WITH SPLICE SLEEVES SHALL HAVE A TENSILE STRENGTH NO LESS THAN 80 PERCENT THE STRENGTH OF THE WIRE BEING SPLICED.

CROSSING DRAWS OR STREAMS
- WHEN CROSSING LANDSCAPE DEPRESSIONS (DRAWS OR SWALES), AND FENCING FOLLOWS THE GROUND SURFACE, LINE POSTS SUBJECT TO UPWARD PULL SHALL BE ANCHORED.
- IF THE FENCE WIRE IS INSTALLED WITH THE TOP WIRE STRAIGHT AND PARALLEL TO THE GROUND SURFACE ON EITHER SIDE OF THE DEPRESSION, EXTRA LENGTH POSTS SHALL BE USED TO ALLOW SPECIFIED DEPTH OF POST PLACEMENT.
- USE SAG BRACING FOR FENCE WIRE STRANDS LOCATED OVER A DEPRESSION TO MAINTAIN REQUIRED SPACING INTERVAL.
- CONSIDER USE OF A SAG BRACE WHEN THE BOTTOM LINE WIRE AT THE LOW SPOT IN CROSSING RISES MORE THAN 20–INCHES ABOVE THE GROUND.

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PROJECT NAME
KALAMAZOO FENCE

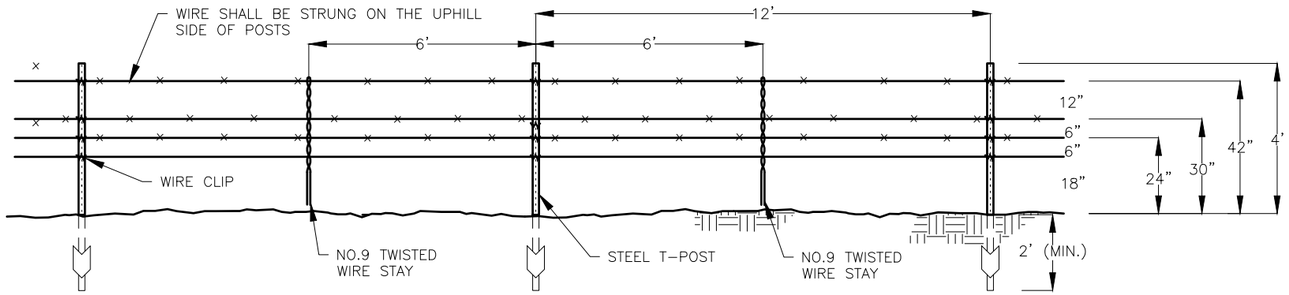
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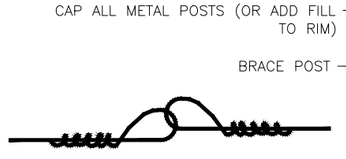
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GENERAL NOTES AND SPECIFICATIONS: KALAMAZOO FENCE

DATE 2/19/2026	ARCHIVE NO.
DESIGNER C.CUBIC	DRAWING SHEET NO. D-2
DRAWN C.CUBIC	
CHECKED K. WILMOT	
PROJECT NO.	SHEET 3 OF 5

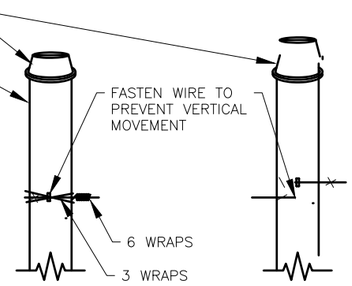
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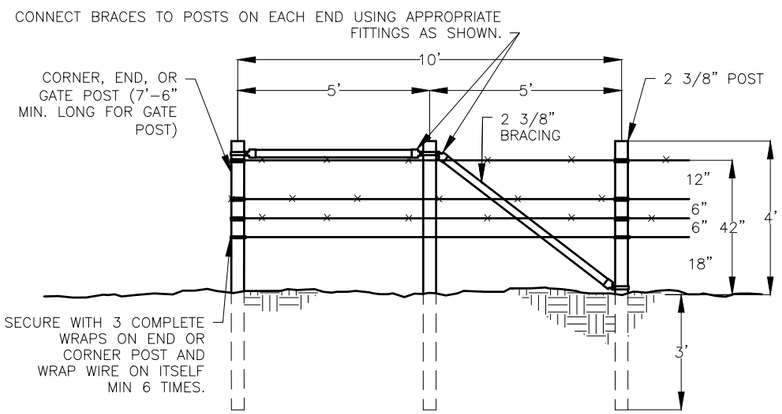
TYPICAL FENCE DETAIL



SPLICING DETAIL

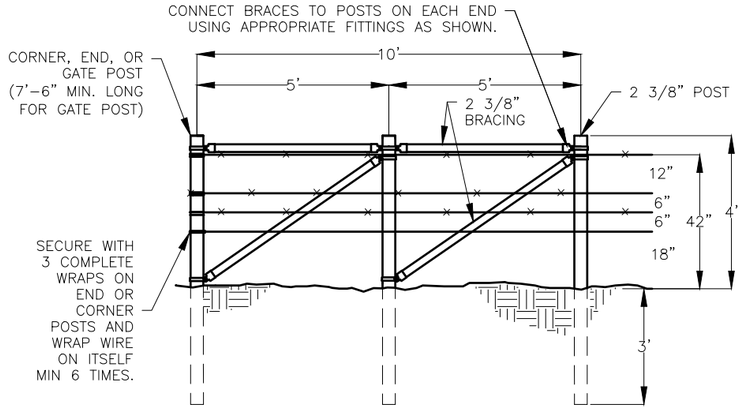


BRACE CONNECTION DETAILS



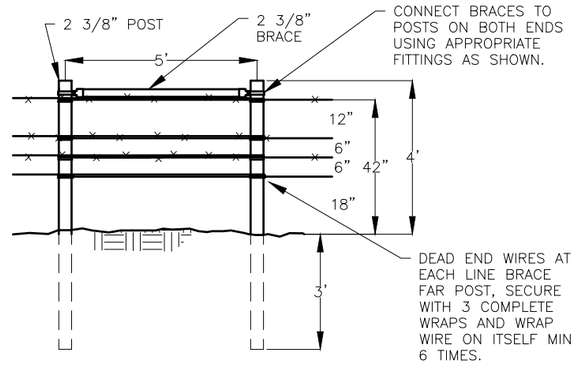
STANDARD BRACE PANEL

FOR RUNS UNDER 660' – EACH END OF GATE, CORNERS, AND TERMINATIONS



HEAVY BRACE PANEL

FOR RUNS UNDER 660' – EACH END OF GATE, CORNERS, AND TERMINATIONS



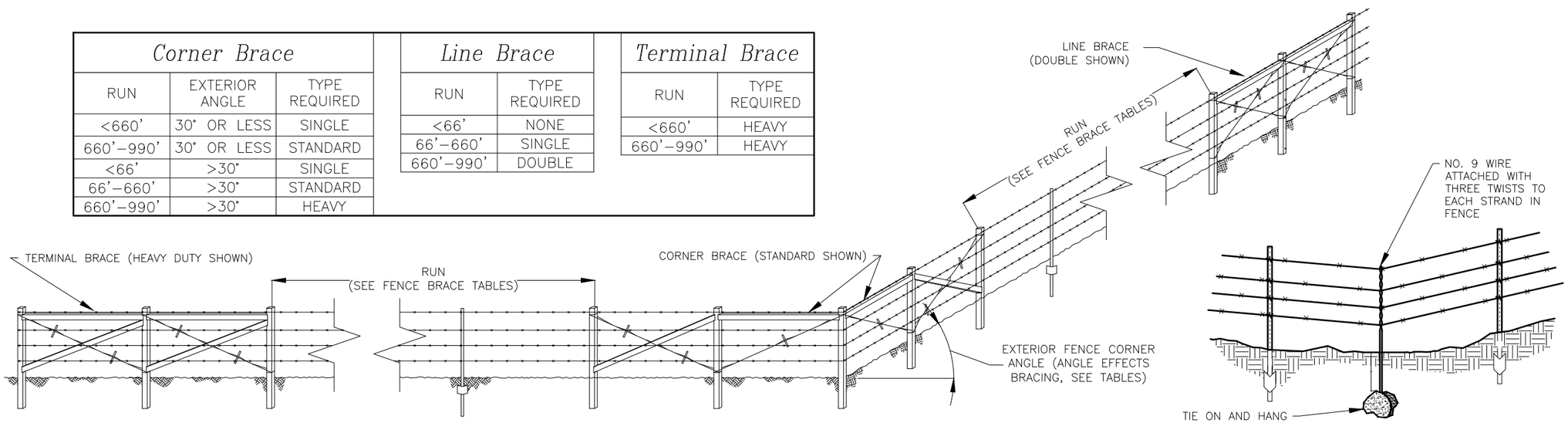
LINE BRACE

NOTE: DOUBLE LINE BRACE REPEATS DETAIL TO A THIRD POST.

Corner Brace		
RUN	EXTERIOR ANGLE	TYPE REQUIRED
<660'	30° OR LESS	SINGLE
660'–990'	30° OR LESS	STANDARD
<66'	>30°	SINGLE
66'–660'	>30°	STANDARD
660'–990'	>30°	HEAVY

Line Brace	
RUN	TYPE REQUIRED
<66'	NONE
66'–660'	SINGLE
660'–990'	DOUBLE

Terminal Brace	
RUN	TYPE REQUIRED
<660'	HEAVY
660'–990'	HEAVY



BRACING PANEL USE DETAIL

NO SCALE

SAG BRACING

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HUMBOLDT-TOIYABE NATIONAL FOREST

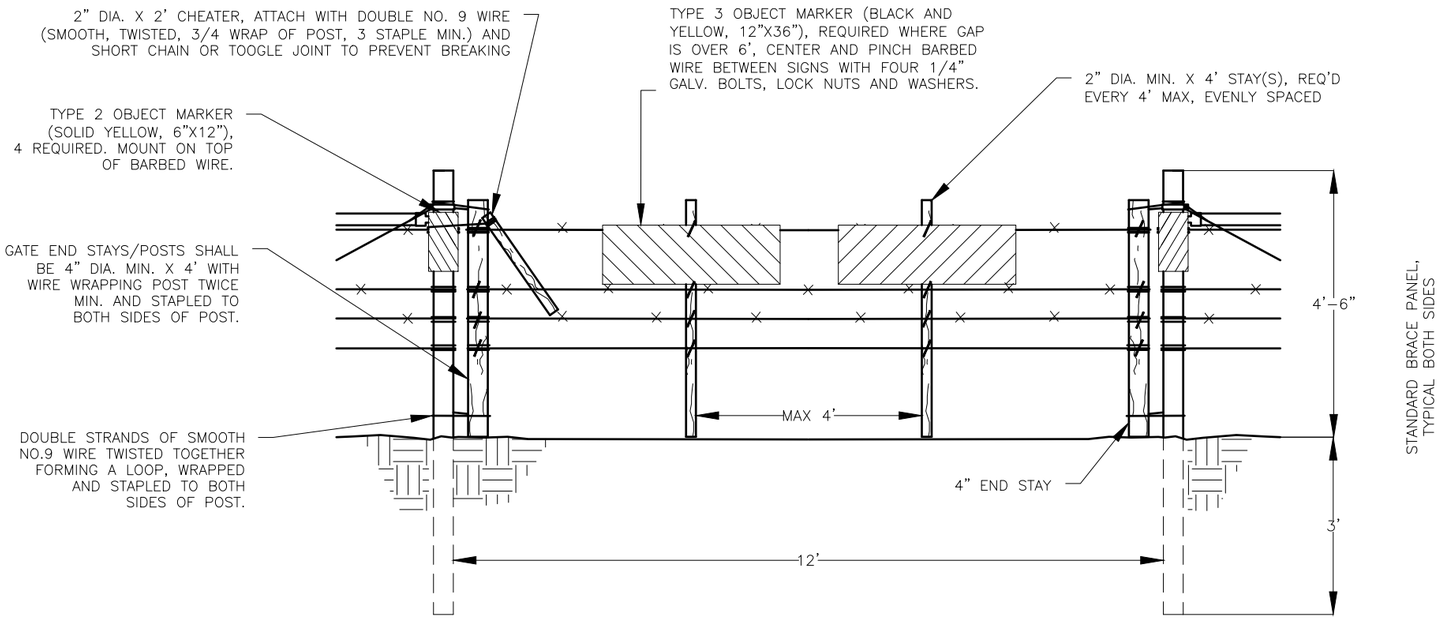
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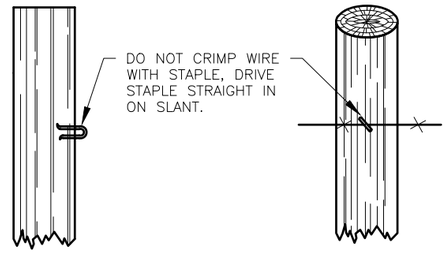
DETAILS:KALAMAZOO FENCE CONSTRUCTION

DATE 4/11/2024	ARCHIVE NO.
DESIGNER C.CUBIC	DRAWING SHEET NO. D-3
DRAWN C.CUBIC	
CHECKED S.JOHNSON	
PROJECT NO.	SHEET 4 OF 5

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WIRE GATE DETAIL



CONSTRUCTION DETAILS

NOTES:

- MATCH WIRE TYPE AND SPACING OF FENCE LINE.
- MATERIALS USED IN CONSTRUCTION OF WIRE GATES SHALL CONFORM TO THE KINDS, GRADES, AND SIZES SPECIFIED FOR A NEW FENCE, AND SHALL INCLUDE NECESSARY FITTINGS AND STAYS.
- WIRE GATE WILL BE MINIMUM 12-FEET WIDE AND CONSTRUCTED USING WOODEN POSTS WITH A WIRE LOOP GATE LATCH. WOOD IS NOT AVAILABLE ON SITE AND WILL NEED TO BE PROVIDED.
- BARBED WIRE SHALL HAVE ONE STAPLE PER WIRE PER POST. STAPLES SHALL BE DRIVEN INTO WOODEN POSTS AS SHOWN ON DRAWINGS AND ANGLED SOMEWHAT TO CROSS THE GRAIN.
- STAPLES SHALL BE NO. 9-GAUGE OR HEAVIER GALVANIZED WIRE WITH A MINIMUM 1 1/2" LENGTH.
- STAPLES SHALL BE DRIVEN INTO POSTS SO THAT THE WIRE WILL NOT MOVE
- WIRE SHALL BE STAPLED AND WIRE TO WOOD STAYS. STAPLES SHALL BE DRIVEN INTO STAYS SO THAT THE WIRE WILL NOT MOVE.
- SMOOTH WIRE SHALL BE USED FOR LOOPS
- TYPE 2 OBJECT MARKERS (6" X 12") SHALL BE INSTALLED ON A GALVANIZED STEEL PLATE.
- TYPE 3 OBJECT MARKERS SHALL BE CONSTRUCTED ON A FIBERGLASS SUBSTRATE AND SHALL BE BLACK AND YELLOW, DOUBLE-SIDED, AND REFLECTORIZED. A "FLEX LITE" FLEXIBLE MARKER MAY BE SUBSTITUTED.
- OBJECT MARKERS SHALL MEET THE REQUIREMENTS OF THE CURRENT MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES (MUTCD) AND ARE INCIDENTAL TO GATE CONSTRUCTION.
- OBJECT MARKERS SHALL BE INSTALLED ON FRONT AND BACK SIDES OF POSTS AND GATE TO BE VISIBLE FROM BOTH DIRECTIONS OF APPROACH.

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DETAILS: KALAMAZOO GATE CONSTRUCTION

DATE 4/11/2024	ARCHIVE NO.
DESIGNER C.CUBIC	DRAWING SHEET NO. D-4
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